

HireIQ

AI-Powered Job Application Automation & Interview Preparation for Pakistan

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What is HireIQ?

HireIQ is an AI-powered web platform that solves four interconnected computer science problems in the Pakistani job market: semantic job-candidate matching using dense vector embeddings, context-grounded interview question generation using Retrieval Augmented Generation (RAG), multi-source heterogeneous data integration across Pakistani job portals, and ATS-compliant CV rewriting with keyword gap analysis. In practical terms, a student uploads their CV once, the platform aggregates relevant jobs from Rozee.pk, Mustakbil, and Indeed Pakistan, ranks them by semantic match score, tailors the CV per application, and then — once the student applies — automatically generates a role-specific interview preparation session using the stored job description and submitted CV. No competing product anywhere connects all three stages — applying, tracking, and preparing — in one system.

Why are we building this?

Pakistan produces 85,000 CS and IT graduates per year yet youth unemployment in the 15–29 age bracket stands at 31% (Pakistan Bureau of Statistics, 2025). The gap is not primarily a skills gap — it is a process gap. Three specific failures motivated this project:

- No job search automation exists for Pakistani portals. Tsenta (YC S26) solves this for US platforms (Workday, Greenhouse) at \$9.99/month. Rozee.pk and Mustakbil — where Pakistani students actually apply — are completely unsupported by any existing tool.
- Interview unpreparedness is the single biggest reason fresh graduates fail their first interviews. University career centers are underfunded. Students go in blind, with no job-specific or company-specific practice.
- The application-to-interview gap is never bridged. Tools that help you apply (Tsenta) and tools that help you prepare for interviews (Final Round AI, Huru.ai) are completely separate products. HireIQ is the first platform where the application feeds directly into the preparation — using the same job description, the same submitted CV, and the same company context.

Pricing is also a direct motivation: all existing interview prep and application tools are priced in USD (\$10–30/month). HireIQ will be priced at PKR 299/month — psychologically and financially accessible for Pakistani students.

How will we build it ?

HireIQ is a web application built on Next.js 14 (frontend and API routes), Supabase (PostgreSQL database, authentication, file storage), and the OpenAI API (embeddings and generation). The four technical components and their feasibility:

- Job Aggregation: Automated scraper using Python and Playwright targeting Rozee.pk and Mustakbil (both server-rendered, low scraping risk). Indeed Pakistan via their official free Publisher API. LinkedIn data seeded manually for demo (documented limitation). Scraper runs every 6 hours via cron on Railway.app free tier.
- Semantic Matching: Student CV converted to vector embedding using OpenAI text-embedding-3-small. Job descriptions stored as embeddings in Supabase using the pgvector extension. Cosine similarity at query time returns a ranked, explainable match score per job.

- CV Tailoring & ATS Scoring: TF-IDF keyword extraction from job description identifies required terms. Gap analysis checks presence (exact + semantic) in the student CV. Constrained rewriting prompt adds missing keywords contextually without fabricating experience.
- Interview Prep Bot (RAG): Job description and submitted CV chunked, embedded, and stored in vector DB at application time. At prep time, relevant chunks retrieved and injected into LLM prompt. Questions generated are grounded in the actual document content — not generic. Answer grading uses a rubric-based evaluation prompt scoring technical accuracy, STAR structure, and keyword coverage.